

Problem #4

Due: Friday, November 22, before midnight. Turn in to canvas, as described in the syllabus.

In this problem set, we will explore a classical chaotic dynamical system that has some relevance for space mission design. An elementary model for the orbits of complex space missions is the circular restricted 3-body problem (CR3BP). This considers the motion of a point mass m_3 in the gravitational fields of two more massive points m_2 and m_1 , with $m_1 > m_2$. The main reference, if you are interested in this topic, is the book by Koon, Lo, Marsden and Ross, which can be found [here](#).

First, the ODEs. We will consider a rotating frame of reference such that m_1 and m_2 , which are moving about their center of mass, are fixed. In this frame of reference, we consider x, y, z to be the three dimensional position of m_3 (the spacecraft) and write down Euler-Lagrange equations (Koon et al Chapter 2):

$$\begin{aligned}\frac{d^2x}{dt^2} &= 2\frac{dy}{dt} - \frac{\partial U}{\partial x} \\ \frac{d^2y}{dt^2} &= -2\frac{dx}{dt} - \frac{\partial U}{\partial y} \\ \frac{d^2z}{dt^2} &= -\frac{\partial U}{\partial z},\end{aligned}\tag{1}$$

where the effective potential, $U(x, y, z)$ is given by

$$U(x, y, z) = -\frac{1}{2}(x^2 + y^2) - \frac{\mu_1}{r_1(x, y, z)} - \frac{\mu_2}{r_2(x, y, z)} - \frac{1}{2}\mu_1\mu_2.\tag{2}$$

Here, $r_1^2 = (x + \mu_2)^2 + y^2 + z^2$ and $r_2^2 = (x - \mu_1)^2 + y^2 + z^2$ are squared distances from m_1 , which is located at $(-\mu_2, 0, 0)$, and m_2 , which is located at $(\mu_1, 0, 0)$, respectively.

Consider a 6-dimensional phase space with variables $X = (x, y, z, dx/dt, dy/dt, dz/dt)$ and the flow described by (1), which will reduce to a first order ODE system for X : $\partial\varphi^t(X)/dt = V(\varphi^t(X))$.

Let us consider $\mu_2 = 0.1$ and $\mu_1 = 1 - \mu_2$, which could be the values when m_1 is Pluto and m_2 is one of its moons.

Question 1: Derive the vector field V , which should give you a nonlinear autonomous dynamical system in $\mathbb{R}^6$. Plot the potential function U on the x - y plane. How many critical points do you see? (3 points)

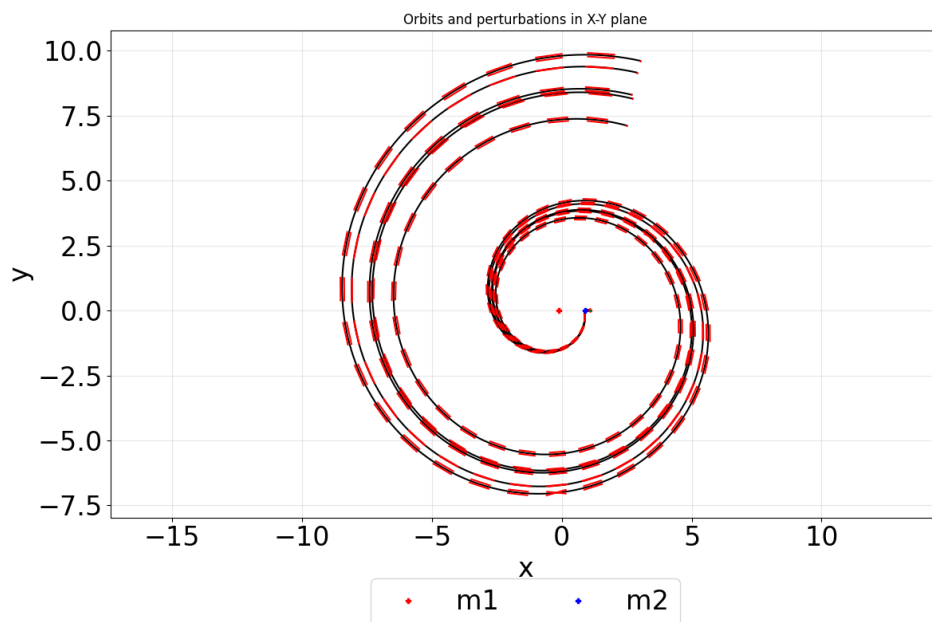


Figure 1: In black, we show orbits of the CR3BP on the x-y plane. The positions of the two masses are shown for $\mu_2 = 0.1$. In red, we show that leading Lyapunov vector computed numerically using power iteration. The initial and final points (at time 10) are shown in green and red respectively for 5 different orbits. The initial points are around L_2

Question 2: We will now work out the physically meaningful fixed points of this system, which are the equilibrium points. That is, these are points where ∇U is the zero vector in $\mathbb{R}^3$, and the gravitation pull of the m_1 and m_2 are perfectly balanced. There are 5 such equilibrium points (see Figure 2.5.1 of Koon et al), three of which are on the x axis ($y = z = 0$). Find (approximately) the two equilibrium points closest to m_2 , on the x axis. (5 points)

Notice that if you used gradient-based approaches or Newton's method, you may miss these equilibrium points. Why is that? (2 points)

These two fixed points, closest to m_2 are called, by convention, L_1 and L_2 . The ' L ' indicates that these are libration (oscillation) points, about which the spacecraft may move. Let the point L_2 have both m_1 and m_2 on the same side.

Question 3: Start close to L_2 and generate orbits. In your answer,

attach orbits from different initial conditions near L_2 on the x - y plane. (2 points). Since the problem has been non-dimensionalized, time 2π , corresponds to one orbital rotation of m_1 and m_2 . You could generate the orbit up to this time or longer. Prove that there is a function (hint: this is the total energy) that remains constant along orbits (2 points).

Such systems are called Hamiltonian systems. This is an example of a system where there is exactly one quantity that remains invariant!

Question 4: Check that the energy is invariant along the orbits you generated. If it is not, explain how the choice of timestep and the time integration scheme affects the invariance (2 points). With Forward Euler, the orbits may look like the black lines in Figure 1.

Question 5: First derive the linearized equations around any fixed orbit, say, $X(t)$. These should be given by

$$\frac{\partial u(t)}{\partial t} = \nabla V(X(t)) u(t), \quad (3)$$

where $u(t)$ is a tangent vector at $X(t)$. The function `tangent` in the file (crudely) approximates the solution of this equation. Using this function or writing your own, answer the following questions (1 point each):

- Is the orbit you picked chaotic? What object in the code tells you the answer?
- If you randomly picked an initial condition, say in a uniform interval on the x -axis around L_2 , would the generated orbit be chaotic?
- What are the other Lyapunov exponents computed by the code?
- What are the true values of the Lyapunov exponents of the system?
- Explain the sources of numerical error in the calculation of the LEs in the code.
- Does this linearized system follow Oseledets theorem?
- Is this system uniformly hyperbolic? Answer this question using the true values of the LEs.
- Do infinitesimal volumes in $\mathbb{R}^6$ contract or expand under the linearized dynamics? How do we know numerically?
- Do infinitesimal volumes in $\mathbb{R}^6$ contract or expand under the original dynamics (flow of V)? How do we know?

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Hint: you may also find the function `plot_les` helpful

Question 6: In the code, `tangent` also computes the Lyapunov vectors along an orbit. The leading Lyapunov vector is shown in red along 5 different orbits in Figure 1. Explain how you would approximate the local unstable manifold using these vectors. (2 points)

Question 7: What are the dimensions of the center manifold and the stable manifold? (1 point) One vector field tangent to the center manifold can be found exactly: what is it? (2 point) Why is it not approximated well by the code? (2 point)

Question 8: True or False: as the length of the orbit goes to infinity, the numerical error in the approximation of LEs and LVs by `tangent` goes to 0. (1 point)

True or False: as the length of the orbit, T , goes to infinity, the variance in our estimates of the LEs is $\mathcal{O}(1/T)$, i.e., follows central limit theorem.

Question 9: Bonus You can solve this question for additional credit. Suppose we would like our spacecraft to go near a target point X_{tar} at time t , and let t be small enough. How can you choose an initial condition $X(0)$ to get near X_{tar} at time t ? (3 points). You are allowed to use all the LVs along any orbit and can approximate the stable manifold locally using the stable LVs. Here small enough time is compared to $1/\lambda_1$, where λ_1 is the leading LE. For context, space missions are about solving such problems. Say you find that $\lambda_1 \approx 1$ in non-dimensional units for this problem. This means that two orbits will diverge on the order e in about 1/6th of the orbital period – so if t is very large, it may not be possible to intervene easily to correct the course of the orbit!